Source Quality

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Source Quality before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Hallucination Failure Types

Without opening the old lesson, explain one key idea from Hallucination Failure Types. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Speech and Audio Systems

Without opening the old lesson, explain one key idea from Speech and Audio Systems. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Prompt Evaluation

Without opening the old lesson, explain one key idea from Prompt Evaluation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Source Quality trace and record the first step you can explain without help.

Use W2: Trace the worked case for Conflicting and time-sensitive sources. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Source Quality explanation against the supplied facts before discussing it.

Use the worked case for Conflicting and time-sensitive sources. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Original allows capacity 14. Signed amendment lowers it to 9 effective day 5. Query concerns days 4 and 6. Trace both and preserve versions.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Source Quality target to a trace and a justified conclusion.

A school notice updated Friday conflicts with a blog quoting Monday's notice about a room change. Which source should you inspect first, and why is newest alone not enough?

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Source Quality case and show the deciding evidence.

An original experiment report and an anonymous unlinked summary describe a method.

Where should verification begin?

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

For a school's current opening time, compare an undated repost with a dated notice issued by the school. Which is the stronger starting source, and what must still be checked?

My independent answer and reason

Write or draw your reasoning and evidence.

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Source Quality adds to Verified Study Buddy and keep the evidence.

Use an approved source list and record why each source fits the learner's question.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A school notice updated Friday conflicts with a blog quoting Monday's notice about a room change. Which source should you inspect first, and why is newest alone not enough?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

W1 Explain the mechanism

Explain the mechanism. Apply the supplied Source Quality facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

W2 Trace the worked case

> Trace the worked case. Apply the supplied Source Quality facts and keep your reasoning.

Notice N says room A from Monday. Signed amendment M references N and changes to B effective Wednesday. Query asks Tuesday, then Thursday. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

W3 Audit the limit

> Audit the limit. Apply the supplied Source Quality facts and keep your reasoning.

A report omits this qualification: An unspecified **authority** or effective-time rule leaves a conflict unresolved; do not invent precedence.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Source Quality facts and keep your reasoning.

Original allows capacity 14. Signed amendment lowers it to 9 effective day 5. Query concerns days 4 and 6. Trace both and preserve versions.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Source Quality facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Source Quality

> A fictional Study Buddy answers which room a club is authorised to use. Explain the labelled

> step before checking the changed condition.

A fictional Study Buddy answers which room a club is authorised to use.

Signed notice A: Hall 1 from day 1. Signed amendment B references A and changes to Hall 3 effective day 8.

An old copy of A is reuploaded on day 10. B remains the authoritative amendment. Queries concern day 6 and day 9.

AI ACADEMY / AI-3/15

Source Quality

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Check authority

B is a signed amendment to A, not an unrelated comment.



2. Day 6

Before effective day 8, authorised room is Hall 1.



3. Day 9

From day 8 onward, authorised room is Hall 3.



4. Ignore upload shortcut

Reuploading old A on day 10 does not revoke B.

CHANGE ONE THING

A signed amendment C explicitly revokes B effective day 12 and restores Hall 1.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Which source fields must be checked before answering day 9?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Store Study Buddy source identity, version, **authority** and effective date, especially for schedules or changing classroom rules.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Verified Study Buddy record.

Project: Verified Study Buddy

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Store Study Buddy source identity, version, **authority** and effective date, especially for schedules or changing classroom rules.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional Study Buddy answers which room a club is authorised to use.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Reuploading old A on day 10 does not revoke B.

1. Day 9

2. Ignore upload shortcut

3. Day 6

4. Check **authority**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Ignore upload shortcut Reuploading old A on day 10 does not revoke B. A source’s upload time and the time its rule takes effect answer different questions.

Preserve version relationships and the query’s intended date. The current authoritative chain gives a consistent answer to both queries.

A later filename or timestamp alone cannot establish that an old record supersedes a signed amendment.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: A signed amendment C explicitly revokes B effective day 12 and restores Hall 1. Which conclusion is supported by the supplied facts?

1. Day 9 stays Hall 3; day 13 becomes Hall 1.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Day 9 stays Hall 3; day 13 becomes Hall 1. The new authoritative change applies from its stated effective date, not retroactively.

Repair a missing building block

Review Hallucination Failure Types (ai-3/14). Goal: Explain the **mechanism** and terminology of generation failures and verification. Short practice: Can we conclude there will definitely be no lunch from S alone?.

Return to this week and explain what you changed.

Review Speech and Audio Systems (ai-3/12). Goal: Explain the **mechanism** and terminology of speech pipelines and multimodal **evidence**.

Short practice: Which stage was initially wrong, and which stage was correctly operating on a wrong **input**?. Return to this week and explain what you changed.

Review Prompt Evaluation (ai-3/9). Goal: Explain the **mechanism** and terminology of controlled prompt experiments. Short practice: Why is B's selection defensible here without calling it universally superior?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

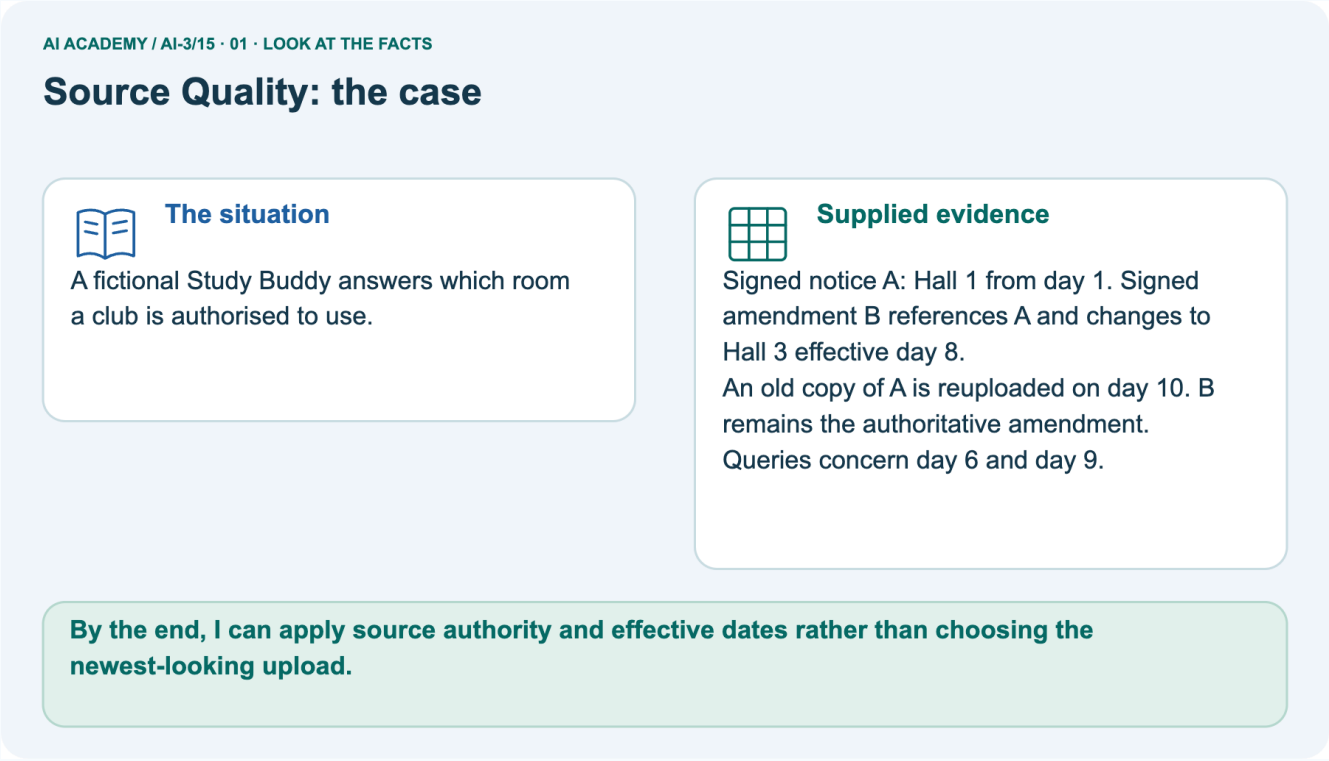
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Source Quality picture from its labels, then record a relevant limit.

01. Source Quality: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

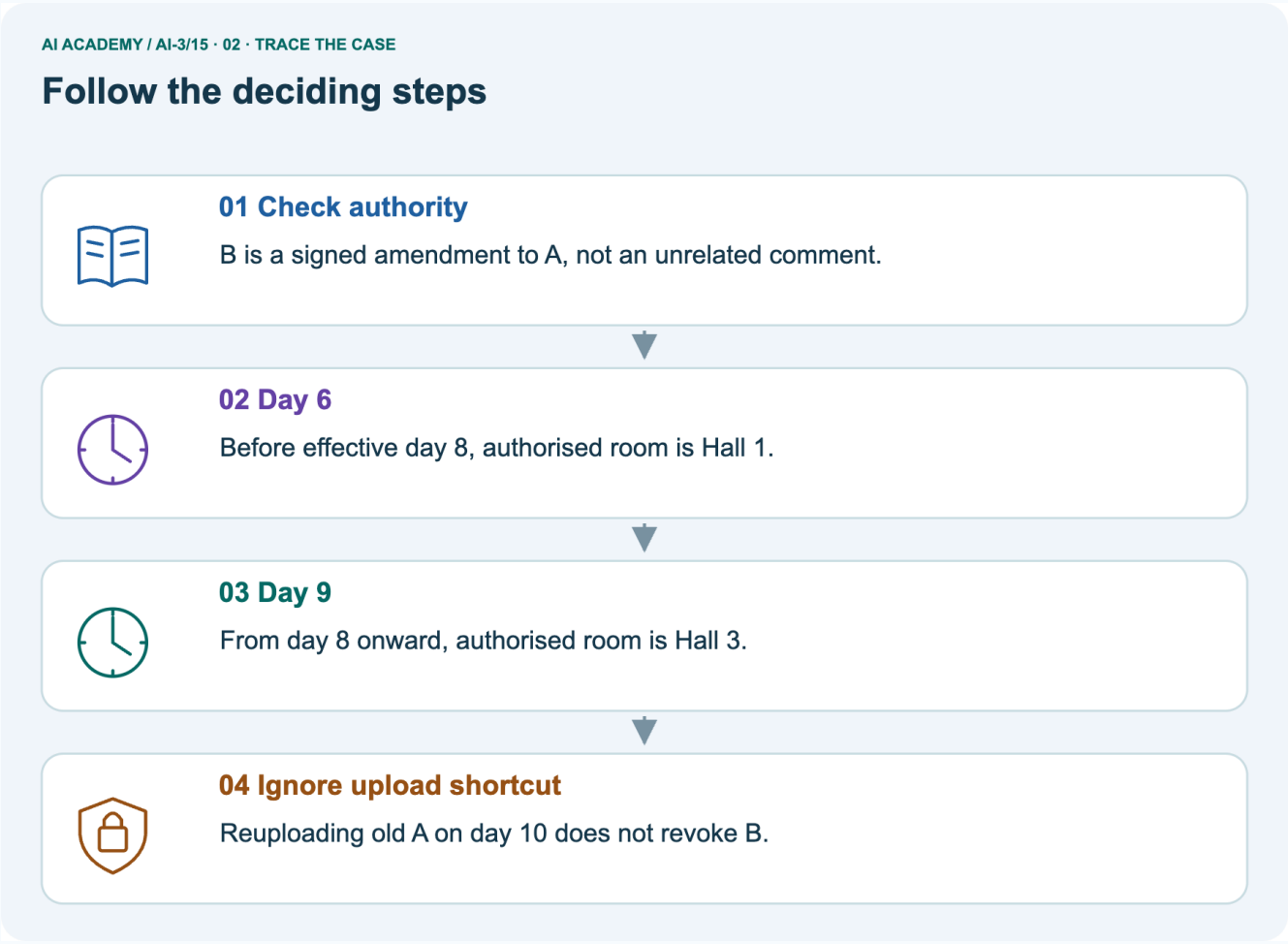
A fictional Study Buddy answers which room a club is authorised to use. Supplied facts: Signed notice A: Hall 1 from day 1. Signed amendment B references A and changes to Hall 3 effective day 8.

An old copy of A is reuploaded on day 10. B remains the authoritative amendment. Queries concern day 6 and day 9.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



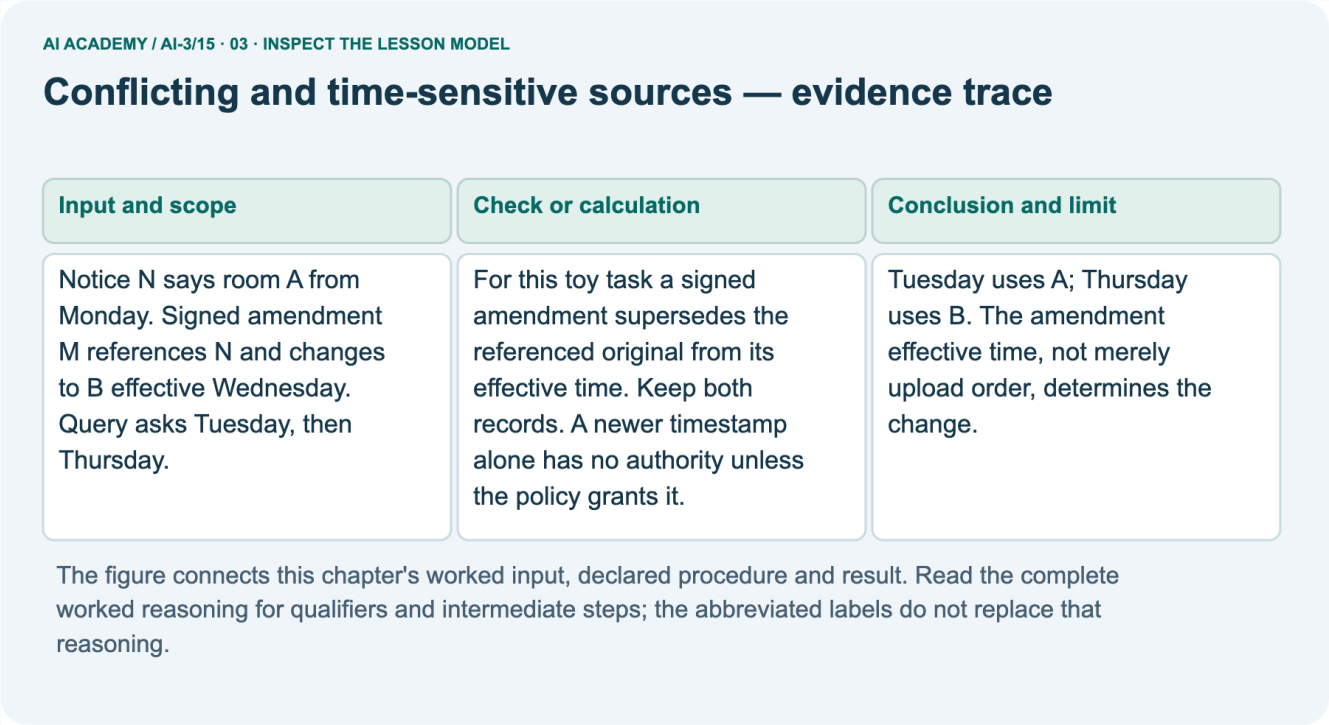
Follow the labelled picture with the matching explanation. Open the vector diagram

Check **authority**: B is a signed amendment to A, not an unrelated comment. → Day 6: Before effective day 8, authorised room is Hall 1. → Day 9: From day 8 onward, authorised room is Hall 3. → Ignore upload shortcut: Reuploading old A on day 10 does not revoke B.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Conflicting and time-sensitive sources — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Conflicting and time-sensitive sources — **evidence** trace. Notice N says room A from Monday. Signed amendment M references N and changes to B effective Wednesday.

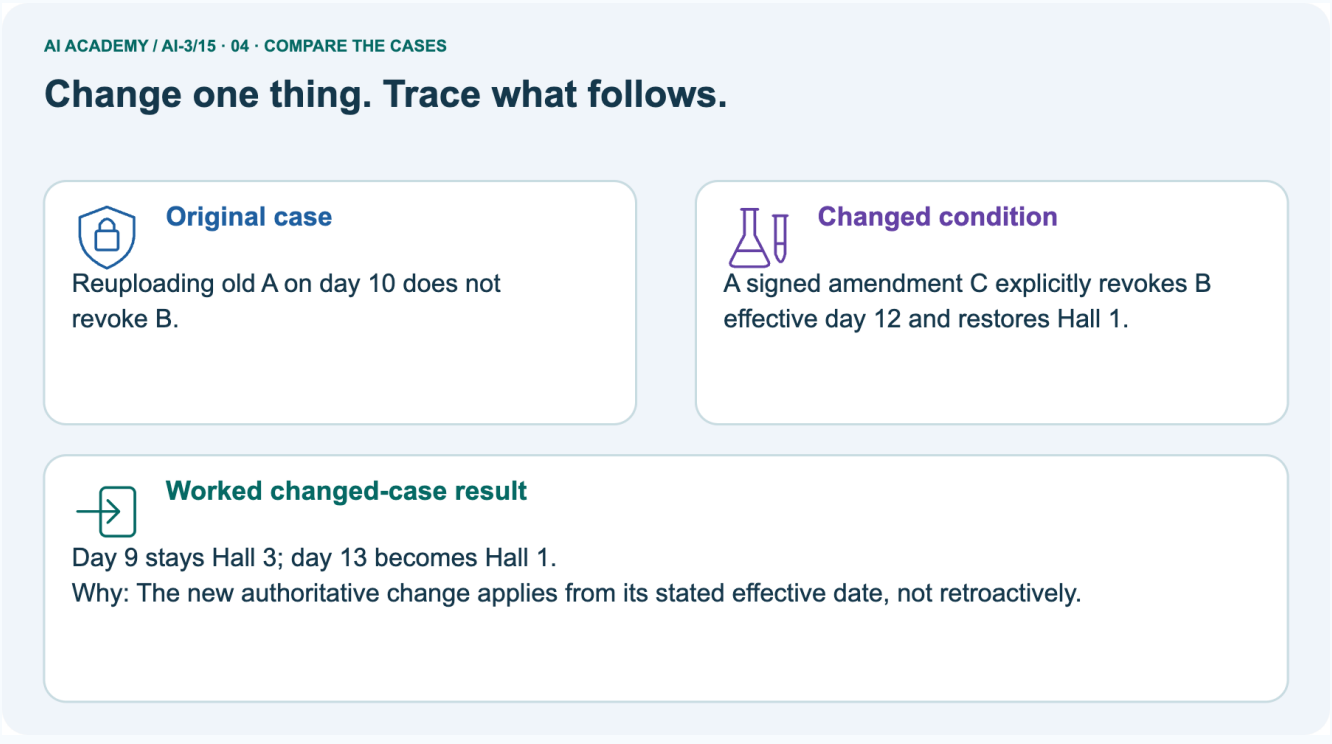
Query asks Tuesday, then Thursday. | For this toy task a signed amendment supersedes the referenced original from its **effective time**. Keep both records.

A newer timestamp alone has no **authority** unless the policy grants it. | Tuesday uses A; Thursday uses B. The amendment **effective time**, not merely upload order, determines the change.

> Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

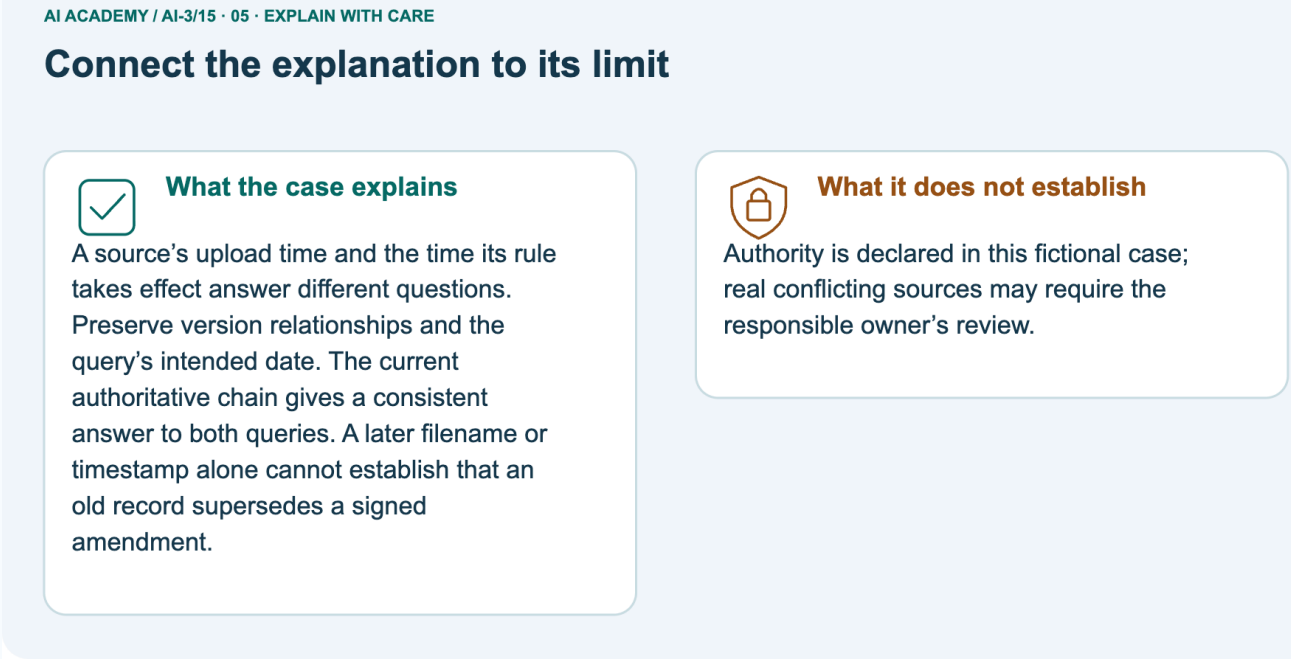
Original case: Reuploading old A on day 10 does not revoke B. Changed condition: A signed amendment C explicitly revokes B effective day 12 and restores Hall 1. Expected result: Day 9 stays Hall 3; day 13 becomes Hall 1.

Reason: The new authoritative change applies from its stated effective date, not retroactively.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

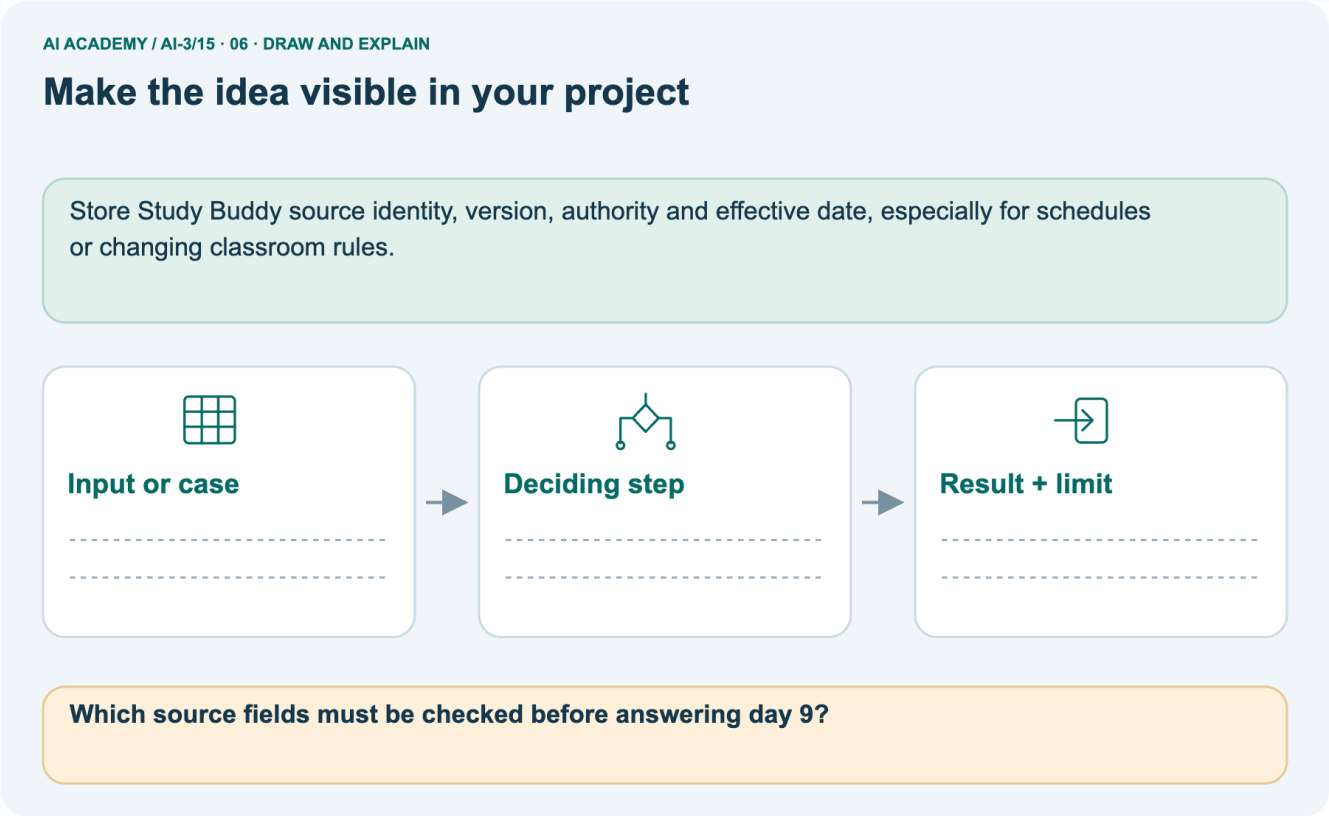
A source’s upload time and the time its rule takes effect answer different questions. Preserve version relationships and the query’s intended date. The current authoritative chain gives a consistent answer to both queries.

A later filename or timestamp alone cannot establish that an old record supersedes a signed amendment. Limit: **Authority** is declared in this fictional case; real conflicting sources may require the responsible owner’s review.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Store Study Buddy source identity, version, **authority** and effective date, especially for schedules or changing classroom rules. Draw your **input**, deciding step, result and limit.

Supported question: Which source fields must be checked before answering day 9?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.